1 A quadratic bound

Definition 1.1. A real number x is nonnegative if $x \geq 0$.

Lemma 1.2. *For every real x, $x^2 \geq 0$.*

Proof. If $x \geq 0$, the product $x \cdot x$ is nonnegative. If $x < 0$, use $x^2 = (-x)^2$. □

Theorem 1.3 (A quadratic bound)**.** *For every $x, a \in \mathbb{R}$, $x^2 - 2ax + a^2 \geq 0$.*

Proof of the quadratic bound. By Lemma 1.2,

$$x^2 - 2ax + a^2 = (x - a)^2 \geq 0.$$ □

Remark. Equality in Theorem 1.3 holds exactly when $x = a$.